

Greyson Ekstrom

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EDUCATION

University of Georgia

B.S. Electrical & Electronics Engineering

Coursework: Circuit Analysis, Digital Logic Design, Intro to Electrical Engineering, Experiential Data Science Specialization

Expected Graduation: May 2029

GPA: 3.84/4.0

LEADERSHIP & INVOLVEMENT

UGA Motorsports Formula SAE

EV Powertrain Lead

Aug. 2026 - Present

- Facilitated discussion between frame, suspension, powertrain, and electronics subteams, following through design and safety requirements to meet document submission deadlines for competition
- Communicated with team members on part selection, design choices, and simulation results to refine electronic models and optimize circuit power consumption for its intended purposes in the system

EV Electronic Circuit Designer

Sep. 2025 - May 2026

- Collaborated among mechanical and electrical teams to develop a printed circuit board using Altium Designer which fulfilled mechanical integration and competition regulations
- Optimized circuit design using LTspice simulation and physical verification before implementing into the electrical system

Robot Dog Electrical Team

Aug. 2025 - May 2026

- Implemented software and electrical communication between motor controllers and onboard Jetson Nano using Linux
- Setup, tested, and troubleshooted persistent software errors and faulty hardware systems during initialization
- Communicated past, present, and future plans for the project and what is needed by the team to achieve later goals

Flying Legion FIRST Robotics

Software Lead

Aug. 2023 - May 2025

- Led a team of 8 developers to design and optimize robotic control software using Java and PhotonVision for software development, delivering the system within a 6-week deadline and surpassing targeted performance markers
- Implemented subsystem control software communicating with brushless motors, encoder feedback loops, PID controls, and servos using Git multibranch workflow to streamline integration and improve subsystem functionality
- Developed autonomous alignment software to optimize robot task completion increasing speed and precision by up to 60%

EXPERIENCE

Warner Robins Best Summer Camp

Summer Camp Coach

May 2026 - Aug. 2026

- Mentored groups of campers, fostering teamwork, leadership, and positive social development in an inclusive environment
- Maintained camper safety, resolved conflicts, and communicated with staff and campers to support a positive experience

PROJECTS

Robotic Arm

Dec. 2025 - Present

- Designed custom gearboxes and mechanical appendages specifically for Nema 17 motors utilizing Autodesk Fusion 360
- Employing Arduino with CNC shield extension running AccelStepper library for multi motor control management
- Integrating Python OpenCV for camera input with future plans on using a Raspberry PI for independent camera vision and correlating reverse kinematics calculations for arm control

Brake System Plausibility Device

Sep. 2025 - Aug. 2026

- Simulated, iterated, and optimized component selection using LTspice and Altium Designer reducing switching time by 20%
- Implemented voltage noise suppression circuitry for both analog and digital components ensuring reliable shutdown behavior

Ready to Move Light

Mar. 2026 - Jun. 2026

- Reviewed and redesigned existing schematics to validate and optimize intended its functionality ultimately fixing critical design oversights and preventing faulty design manufacturing delay
- Developed a custom circuit board which detects greater than 60 volts within the high voltage system to indicate active voltage level by monostably flashing a light with a 555 timer

SKILLS

Design & Simulation: Altium Designer, KiCad, LTspice, PSpice, Multisim

Programming: Python, MATLAB, Verilog, Java

Operating Systems: Linux, Windows 10 & 11

Hardware: Jetson Nano, Raspberry PI, Arduino

AWARDS

University of Georgia Dean's List

Dec. 2025, May 2026